

Mavuno: offline coffee-leaf triage for smallholder farmers

Solo Builder (Najmitdin) · World Bank × Hack-Nation · Small AI for Development, Track B: Agriculture · github.com/mukhitdinov0107/mavuno · mukhitdinov0107.github.io/mavuno

Challenge tackled

Noor, a smallholder coffee farmer in central Kenya, sees her harvest drop and cannot tell why; the extension officer visits twice a year. Kenya's coffee yields fell from 973 to 474 kg/ha (1963/64 to 2021/22) across 800,000+ smallholders, and one extension agent serves about 1,380 farmers against FAO's recommended 400 (Kenya Ministry of Agriculture, *Coffee Strategy 2024–2029; Extension Manual*, 2025). **Users:** Noor and her 16-year-old daughter, sharing one low-end Android phone with no Wi-Fi; Swahili speakers with low screen literacy.

How it works. 5 leaf photos from 3 trees → quality check → on-device leaf model → keep photos it is ≥70% sure of → combined with 8 picture answers and plot data as clues → a transparent table scores 11 causes → likely cause + 2 reasons + vetted advice card, or "not sure, ask the extension officer" → consent → encrypted report queued for the cooperative.

Tools / ML models used

- **MobileNetV3-Small** (ImageNet weights, fine-tuned in TensorFlow/Keras): leaf classification, 6 classes; int8 weights, 1.1 MB, LiteRT on the phone, ~50 ms/photo
- **Rule-based fusion** (pure Kotlin): expert-weighted evidence table, softmax, 4 abstain rules
- **Android:** Kotlin, Jetpack Compose, CameraX, Room + SQLCipher, Keystore, WorkManager
- **Web demo:** LiteRT.js running the same model + JS port of the scoring
- **Data:** JMuBEN/JMuBEN2 (Kenya), BRACOL (Brazil), PlantDoc; all CC BY 4.0

What worked well

Kenyan leaf crops (held out)	97.5%
Whole leaves, another country (BRACOL)	78.3%
...photos it is confident about (71% of them)	88.1%
Non-coffee leaves rejected	99.6%
Fusion scenarios passing (incl. every abstain rule)	20/20
APK size (limit 40 MB)	13 MB

Full offline check in airplane mode on a 2 GB-RAM emulator; Swahili and English; every on-screen word comes from a reviewed list (enforced by tests).

Limits and next 24 hours

Untested on real Kenyan field photos (hands, soil, shadows) and local varieties; weak on leaf miner; scoring weights and advice cards await agronomist review. **With 24 more hours:** the CHIRPS/iSDAsoil rain and soil pack, the cooperative officer dashboard, recorded Swahili audio, and 200 field photos from Nyeri to re-test the model.

Prototype only; not for real farm decisions. Code MIT; data CC BY 4.0 (see docs/data_card.md).

What was challenging

- **The dataset was smaller than it looked.** JMuBEN's 18,984 "healthy" files are ~17 distinct leaf crops saved many times rotated or flipped. A random split would leak copies into the test set. Fix: rotation-invariant hashing, split by distinct leaf, each leaf counted once in tests.
- **Quantization broke the model.** Full int8 dropped accuracy 95% → 54%. Fix: int8 weights with float activations (93%, still 1.1 MB).
- **App ≠ Python.** An on-device parity test found single-step resizing of large photos shifted confidence 0.69 → 0.42. Fix: stepwise downscaling (gap now ≤0.03).
- **BRACOL's published zip is truncated;** streamed extraction recovered 1,402 of 1,747 images.

How we spent our time (Sat–Sun, ET)

- 20:00–20:30 · PRD, scoring engine, 20 test scenarios, Swahili/English packs
- 20:30–22:30 · Android app, emulator testing, redesign for low literacy
- 22:30–00:00 · Datasets, duplicate audit, training, quantization, parity test
- 00:00–02:00 · Docs, public repo + APK, browser demo on GitHub Pages
- 02:00– · Videos, submission